Roadside Accident Detection System

RADS

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AI-Powered Real-Time Accident Detection using YOLOv8

Academic Year 2024-2025

Problem Statement

Road Accident Crisis

Global road accidents cause millions of deaths annually, with delayed emergency response being a critical factor.

Manual Monitoring Limitations

Traffic control rooms rely on human operators watching multiple CCTV feeds simultaneously, leading to fatigue, missed incidents, and slow response times.

The Solution

An automated AI system that detects accidents from video footage and instantly alerts authorities with location data.

System Overview

YOLOv8 Detection

Fine-tuned model on 6,424 annotated frames (4 vehicle classes)

5-Rule Analysis

Velocity drop, IoU overlap, area change, abnormal stop, trajectory deviation

Web Dashboard

PHP + MySQL platform for operators to upload videos and manage alerts

SMS Alerts

Fast2SMS integration for instant emergency notifications

Technical Architecture

Presentation Layer
HTML/CSS/JavaScript Dashboard

Application Layer
PHP Backend (Auth, Upload, Analysis)

AI Detection Engine
Python + YOLOv8 + 5-Rule System

Data Layer
MySQL (operators, logs, alerts)

Technology Stack

- **YOLOv8n (Ultralytics)**
- PyTorch 2.7.1 + CUDA 12.4
- OpenCV + NumPy
- PHP 8 + MySQL
- XAMPP Server
- Fast2SMS API
- NVIDIA GTX 1660 Super (6GB)

5-Rule Accident Detection

1. **Velocity Drop** — 60% deceleration from rolling avg — 35%
2. **IoU Overlap** — >0.18 for 3 consecutive frames — 30%
3. **Area Change** — >50% sudden box area change — 10%
4. **Abnormal Stop** — Fast → Stopped transition — 20%
5. **Trajectory Deviation** — >125° direction change — 5%

Final Threshold

≥ 0.65

Temporal Smoothing
4-frame persistence

Multi-Rule Bonus
20% boost for 3+ rules

Dataset & Model Training

Dataset Specification

Training Images	**6,424**
Validation Images	**1,606**
Classes	**4 (car, bike, bus, truck)**
Annotation Tool	**CVAT**
Format	**YOLOv8 (YOLO format)**
Resolution	**640 × 640 pixels**

Training Configuration

Base Model
YOLOv8n (pre-trained on COCO)

Epochs: 50 (patience: 10)
Batch Size: 8
Optimizer: SGD (momentum: 0.937)
GPU: NVIDIA GTX 1660 Super

Final mAP50

81.2%

Web Platform Features

Operator Registration

Location-based user accounts

Video Upload

Drag & drop MP4/AVI/MOV

Real-time Detection

3 fps processing, live status

Confidence Display

Animated score bar 0-100%

SMS Alerts

Fast2SMS integration

Detection Logs

Full audit trail in MySQL

Performance Metrics

81.2%
mAP50

73.0%
Precision

73.0%
Recall

Class-wise Performance

Class	Precision	Recall	mAP50
Car	0.78	0.74	0.83
Motorbike	0.69	0.70	0.78
Bus	0.74	0.72	0.81
Truck	0.71	0.73	0.82

Challenges & Solutions

⚠ **False Positive Rate**	✓ Raised threshold to 0.65, temporal persistence (4 frames), multi-rule requirement (3+ rules)
⚠ **Processing Speead**	✓ Reduced sampling to 3 fps, model fusion, 150-frame cap → 40-50% faster
⚠ **Live CCTV Access**	✓ Upload-based workflow with static location assignment per operator

Future Enhancements

1. Live RTSP Stream Integration
Real-time processing of CCTV feeds instead of uploaded videos

2. Dedicated Accident Classifier
Binary CNN model trained on labeled accident clips for higher accuracy

3. Multi-Camera Dashboard
Simultaneous monitoring of multiple locations with unified alert panel

4. Severity Classification
Classify accidents as minor/moderate/critical to prioritize response

5. Edge Deployment
Deploy on NVIDIA Jetson for on-site processing without video upload

Conclusion

✓ Successfully fine-tuned YOLOv8n achieving 81.2% mAP50

✓ Developed robust 5-rule detection system with 0.65 confidence threshold

✓ Built complete web platform with automated SMS alerting

✓ Optimized processing speed by 40-50% while reducing false positives

Bridging AI and Emergency Response

THANK YOU

Questions?